Affine Models and Calibration to data



Project in Financial Engineering
15th of May, 2026.

Emil Beck Aagaard Korneliussen

Supervisors:
Thomas Kokholm
Elisa Nicolato

Department of Economics
Aarhus University

Abstract (English)

This project studies...

Resumé (Dansk)

Dette projekt undersøger...

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1 Introduction

A high level overview of why we care as well as what we seek to answer. In particular a more fluffy description of what an AJD model is and the usefullness of the extended transform.

$$\phi^X(v, u, X_t, t, T) = \mathbb{E}\left[e^{-\int_t^T R(X_s, s)\, ds} (v_0 + v_1 \cdot X_T) e^{u \cdot X_T} \mid \mathcal{F}_t \right] \tag{1.1}$$

2 Affine Models, its Extended Transform & Fourier Pricing

In this chapter we introduce the Affine-Jump-Diffusion model (AJD), for this purpose we follow (Duffie et al., 2000), who were the first to synthesize the litterature in the area.

Consider a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions. Suppose $X = (X_t)_{t\geq 0}$ is a n-dimensional Markov process, with respect to $(\mathcal{F}_t)_{t\geq 0}$ with state space $D \subset \mathbb{R}^n$. Suppose X solves the stochastic differential equation (SDE):

$$\mathrm{d}X_t = \mu(X_t)\ \mathrm{d}t + \sigma(X_t)\ \mathrm{d}W_t + \ \mathrm{d}Z_t,$$

where $W = (W_t)_{t\geq 0}$ is a standard $\mathcal{F}_t$-Brownian motion (BM), $\mu : D \to \mathbb{R}^n$, $\sigma : D \to \mathbb{R}^{n\times n}$ and $Z = (Z_t)_{t\geq 0}$ is a pure jump process with fixed jump distribution ν on $\mathbb{R}^n$ and arrival intensity $\lambda = (\lambda(X_t))_{\geq 0}$, for some $\lambda : D \to \mathbb{R}_{\geq 0}$.

Assume that X_0 is *known*[1]. Let $R : D \to \mathbb{R}$ denote the short rate. The AJD model assumes that μ, $\sigma\sigma^\top$, λ and R are affine with their dependence determined by the coefficients (K, H, l, ρ) defined by

- $\mu(x) = K_0 + K_1 x$, for $K = (K_0, K_1) \in \mathbb{R}^n \times \mathbb{R}^{n\times n}$.
- $(\sigma(x)\sigma(x)^\top)_{ij} = (H_0)_{ij} + (H_1)_{ij} \cdot x$, for $H = (H_0, H_1) \in \mathbb{R}^{n\times n} \times \mathbb{R}^{n\times n\times n}$.
- $\lambda(x) = l_0 + l_1 \cdot x$, for $l = (l_0, l_1) \in \mathbb{R} \times \mathbb{R}^n$.
- $R(x) = \rho_0 + \rho_1 \cdot x$, for $\rho = (\rho_0, \rho_1) \in \mathbb{R} \times \mathbb{R}^n$.

Furthermore, for $c \in \mathbb{C}^n$ we define the *jump transform* by $\theta(c) = \int_{\mathbb{R}^n} \mathrm{e}^{c\cdot z}\ \mathrm{d}\nu(z)$ whenever this integral is well defined. The jump transform determines the distribution of the jump-sizes. Indeed, for any $u \in \mathbb{R}^n$, set $c = iu$ then

$$\theta(iu) = \int_{\mathbb{R}^n} \mathrm{e}^{iu\cdot z}\ \mathrm{d}\nu(z) = \phi_\nu(u),$$

where ϕ_ν denotes the characteristic function of ν. Since a probability measure is uniquely determined by its characteristic function, the jump transform uniquely determines ν.

A *characteristic* is a tuple containing the coefficients of X as well as the short rate R, that is, $\chi = (K, H, l, \theta, \rho)$. Given an initial condition X_0 a characteristic completely determines the distribution of X as well as the effects of discounting. Thus, χ also specifies a transform (when well defined) of X_T conditional of $\mathcal{F}_t$ for $t \leq T$ given by:

$$\begin{aligned}\psi^\chi : \mathbb{C}^n \times D \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} &\to \mathbb{C},\\ (u, X_t, t, T) &\mapsto \mathbb{E}_t^\chi\left[\mathrm{e}^{-\int_t^T R(X_s)\ \mathrm{d}s}\mathrm{e}^{u\cdot X_T}\right].\end{aligned}$$

Where $\mathbb{E}_t^\chi = \mathbb{E}^\chi[\cdot \mid \mathcal{F}_t]$ denotes the expectation under the distribution of X determined by χ conditional on $\mathcal{F}_t$. We now prove the following result, which is also shown in (Duffie et al., 2000):

Theorem 2.1
Assume the characteristic χ is *well-behaved* at $(u, T) \in \mathbb{C}^n \times \mathbb{R}_{\geq 0}$ as defined in Definition 1 in (Duffie et al., 2000). Then,

$$\psi^X(u, x, t, T) = \mathbb{E}_t^\chi\left[\mathrm{e}^{\int_t^T R(X_s)\ \mathrm{d}s}\mathrm{e}^{u\cdot X_T} \mid X_t = x\right] = \mathrm{e}^{\alpha(t)+\beta(t)\cdot x},$$

where α and β are the (unique) solutions to the complex-valued ODEs:

$$\begin{aligned}\frac{\partial \beta}{\partial t}(t) &= \rho_1 + K_1^\top \beta(t) - \frac{1}{2}H_1\beta(t) - l_1(\theta(\beta(t)) - 1)\\ \frac{\partial \alpha}{\partial t}(t) &= \rho_0 - K_0 \cdot \beta(t) - \frac{1}{2}\beta(t)^\top H_0 \beta(t) - l_0(\theta(\beta(t)) - 1),\end{aligned}$$

with boundary conditions $\beta(T) = u$ and $\alpha(T) = 0$.

[1] In the sense that its distribution is degenerate at a known value x_0, i.e., $\mathbb{P}(X_0 \in A) = \delta_{x_0}(A)$ for any measurable set A, where δ_{x_0} denotes the Dirac measure at x_0.

Proof

Q.E.D.

This result is very nice because besides some regularity conditions we now have an explicit way of finding the extended transform of ψ^X. The as mentioned in Chapter 1 the extended transform serves multiple use-cases. In particular, the characteristic function of any (*well-behaved*) ADJ model is now obtainable, either in closed form or through numerical estimation.

2.1 Fourier Pricing

References

Duffie, D., Pan, J., & Singleton, K. (2000). Transform analysis and asset pricing for affine jump-diffusions. *Econometrica*, *68*(6), 1343–1376.